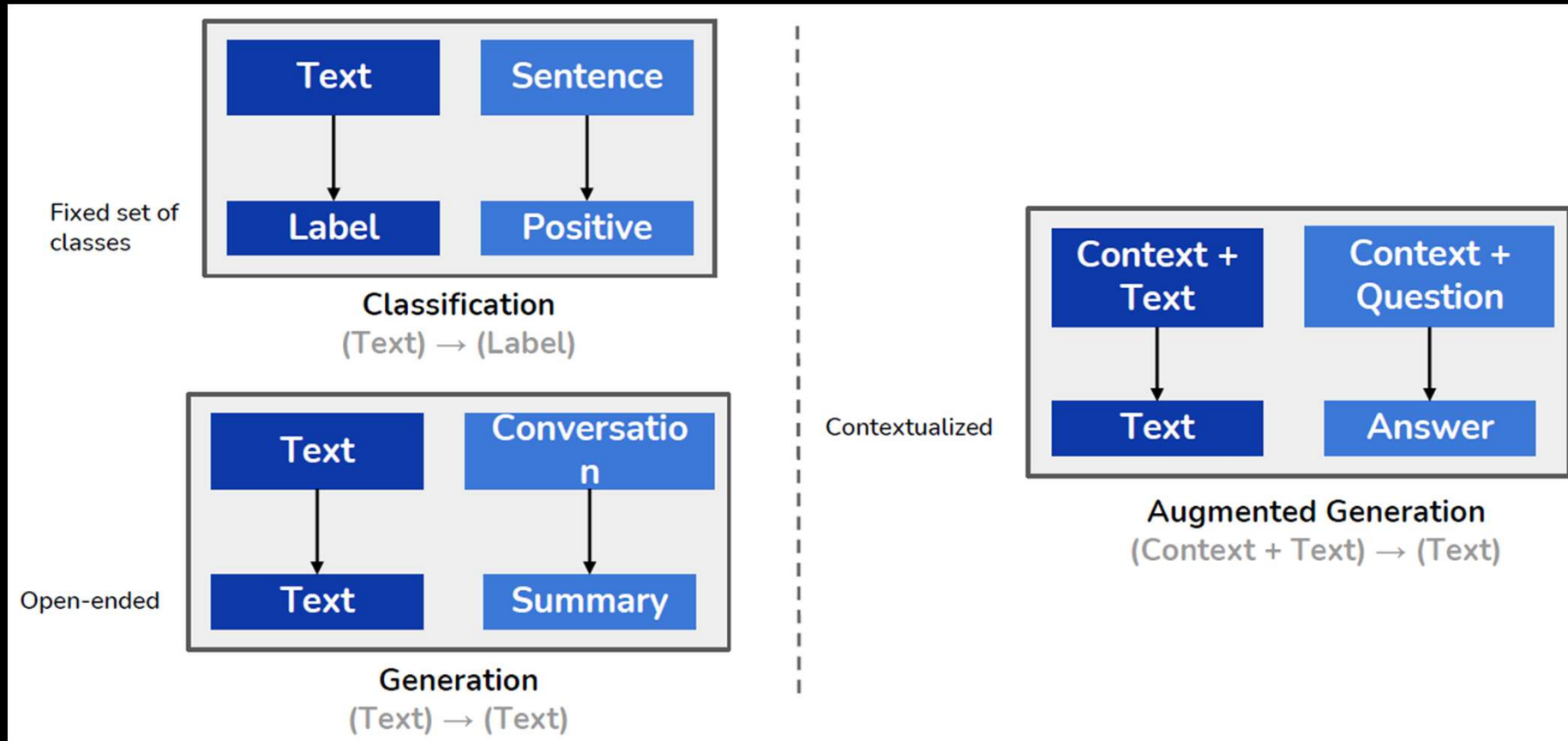




Generative AI Academy

Business problems solved by LLMs – a taxonomy



Accessing LLMs through Azure Open AI

Azure provides controlled, fast access to a host of Open AI LLMs.

Base models

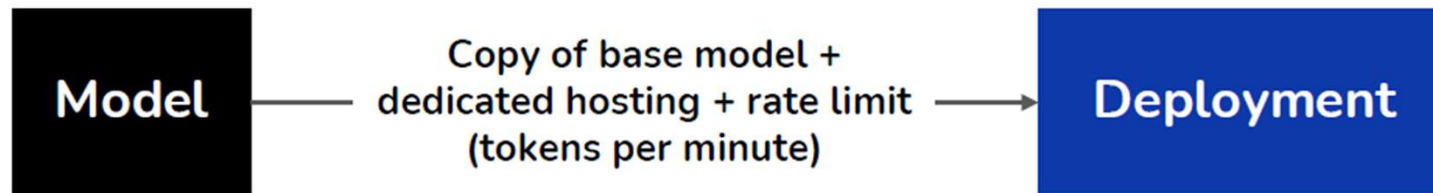
 Deploy  Create a custom model  Column options  Refresh

 Search

Model name ▾	Model version ▾	Created at ▾	Status ▾	Deployable ▾
gpt-35-turbo	0613	6/19/2023 5:30 AM	✓ Succeeded	✓ Yes
gpt-35-turbo	0301	3/9/2023 5:30 AM	✓ Succeeded	✓ Yes
gpt-35-turbo-16k	0613	6/19/2023 5:30 AM	✓ Succeeded	✓ Yes
text-embedding-ada-002	2	4/3/2023 5:30 AM	✓ Succeeded	✓ Yes
text-embedding-ada-002	1	2/2/2023 5:30 AM	✓ Succeeded	✓ Yes

Accessing LLMs through Azure Open AI

Azure provides controlled, fast access to a host of Open AI LLMs.



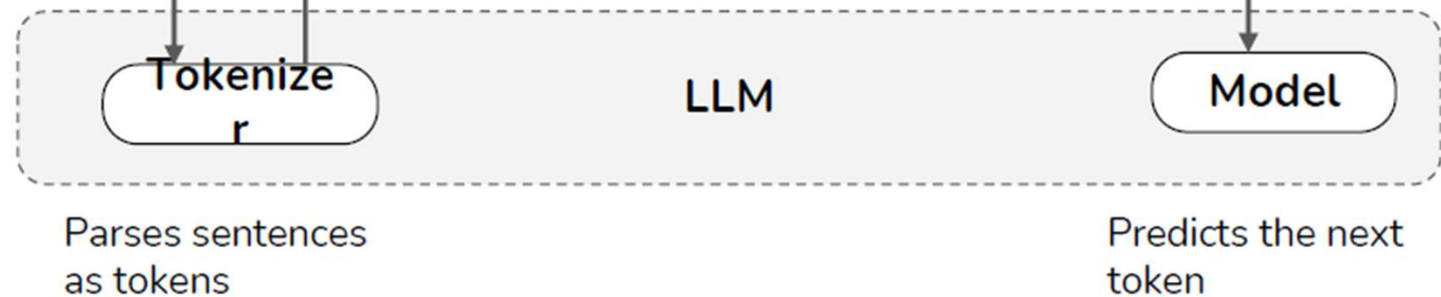
Deployment name ▾	Model version ▾	Capacity	Model deprecati... ▾	Content Filter ▾	Rate limit (Tokens per minute) ▾
☒ gpt-35-turbo	0613	134K TPM	7/5/2024	Default	134000
☐ gpt-35-turbo-16k	0613	135K TPM	1/15/2024	Default	135000
☐ text-embedding-ada-002	2	133K TPM	2/2/2025	Default	133000

Understanding tokens

A token refers to a segment or piece of text, such as a word, punctuation, or other meaningful element, into which input text is divided for processing by the model.

The movie was awesome.
Overall, the experience
was positive.

[The,movie,was,awesome,.,Overall,,the
,experience,was,positive,.]



Demo: tokenizer: <https://platform.openai.com/tokenizer>

Azure Open AI Playground

The Azure Open AI Playground enables iterative development for prompt engineering, that is, designing specific instructions for LLMs to accomplish a task.

The screenshot displays the Azure Open AI Playground interface. At the top, the breadcrumb "Azure OpenAI Studio > Chat playground" is visible, along with a "Privacy & cookies" link and a "Deploy to" button. Below the breadcrumb, the title "Chat playground" is shown, followed by links for "Import setup", "Export setup", and "Show panels".

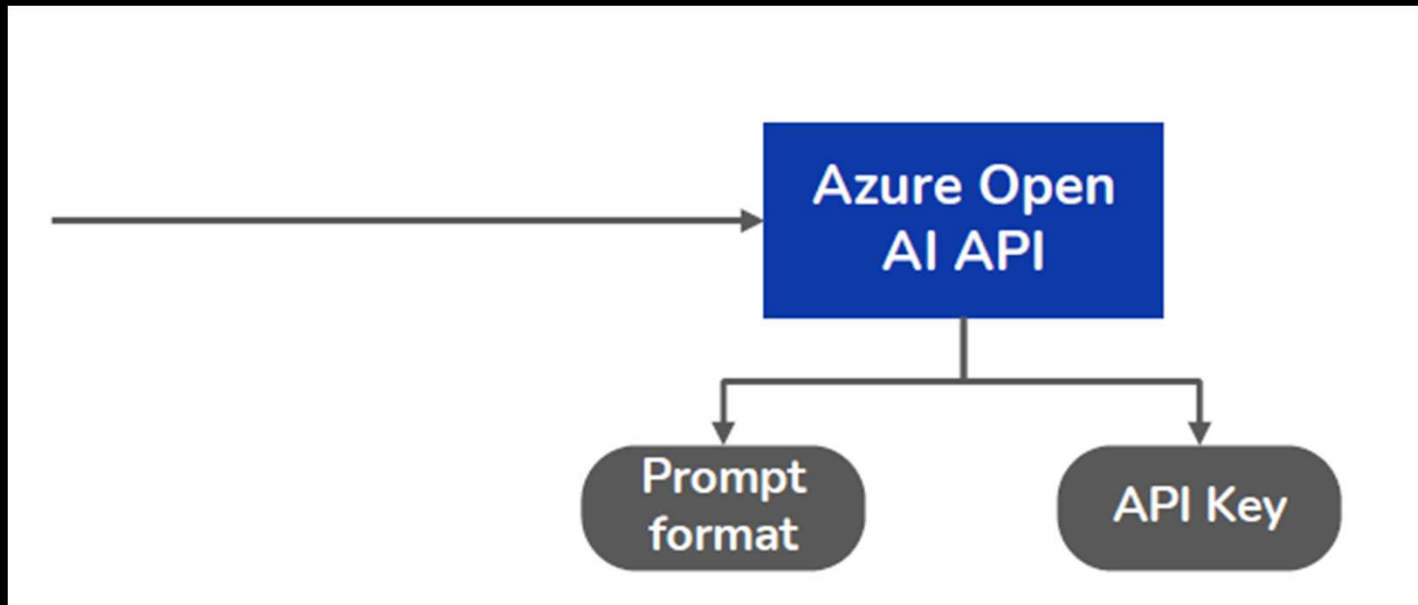
The interface is divided into three main sections:

- Setup (Left Panel):** Contains an "Apply changes" button, a "Use a system message template" section with a "Select a template" dropdown, a "System message" section with a text area containing "Assign sentiment to the product review presented in the user input. Answer in one word.", and an "Examples" section with a "Using examples" sub-section and an "Add" button.
- Chat Area (Center):** Features a "Clear chat" button, "Playground settings", "View code", and a "Show JSON" toggle. The chat history shows a system message (blue bubble) and a user query (white bubble). The system message reads: "Excellent audio quality esp. with an Apple product. Spatial Audio is extraordinary; try with any Spatial Audio track on Apple Music and you will be amazed (I didn't try with any other Dolby Atmos streams). It has all the advertised features (head tracking, etc). more so, if used with Apple sound sources. Battery life is just ok and it comes for around six hours with Dolby Atmos playback; it may be more with normal music though. I found the battery to be draining faster when used with online meetings on PC or on phones. It pairs fine with Android phones and Windows 11 devices and can use basic features like any other Bluetooth earbuds." The user query is "Positive".
- Configuration (Right Panel):** Includes a "Deployment" section with a "gpt-35-turbo" dropdown, a "Session settings" section with a "Past messages included" slider set to 1, and an "Input tokens progress indicator" showing 18/4000.

At the bottom of the chat area, a status bar indicates "18/4000 tokens to be sent" with a send button.

Accessing Azure Open AI via APIs

- Colab notebook
- Python or any other application



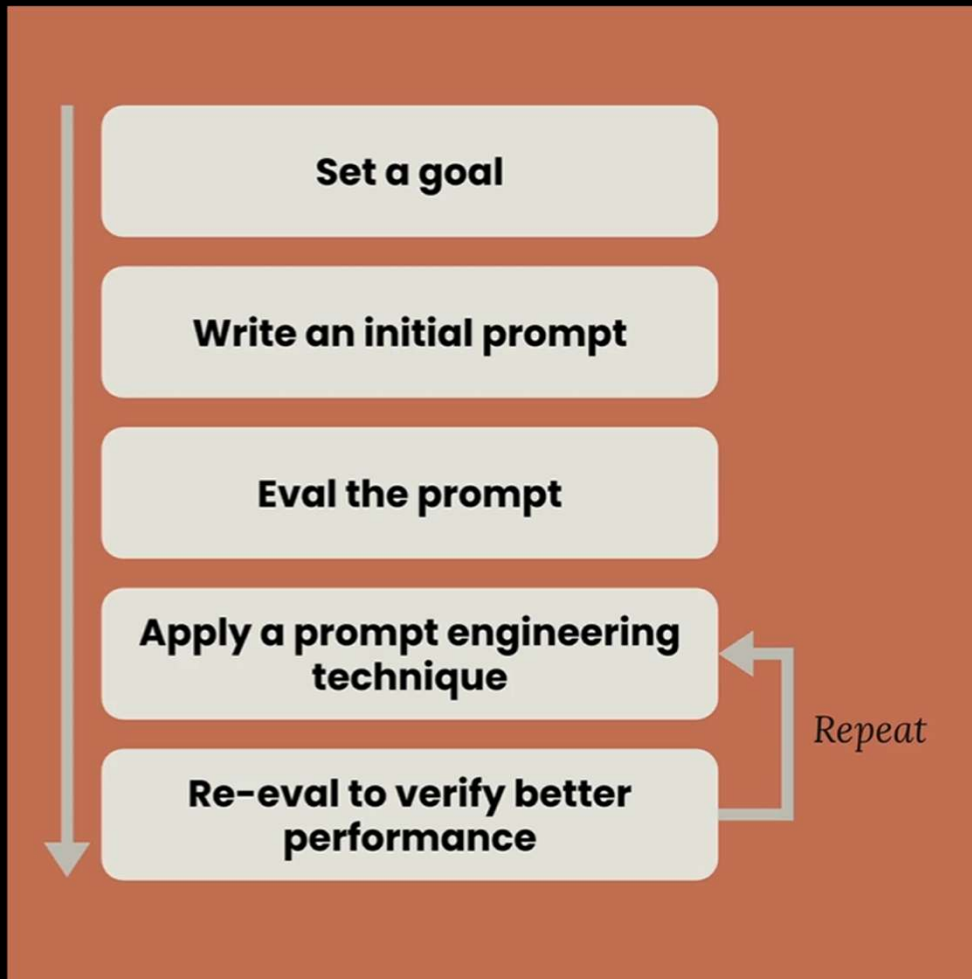
Accessing Azure Open AI via APIs

Demo and hands-on

- Accessing Azure OpenAI from Colab notebook
- Accessing Azure Open AI from your IDE
(VS Code or PyCharm or Spyder or any other)

Prompt Engineering

Prompt Engineering



Prompt Engineering

Goal:

Write a prompt that generates a 1-day meal plan for an athlete based upon their height, weight, goal, and dietary restrictions

Prompt Engineering

Sample Input

Height	180cm
Weight	75kg
Goal	Maintain weight and improve power
Dietary restrictions	Lactose intolerance



Our Prompt



Ideal Output

Here is a one-day meal plan for an athlete aiming to maintain weight and improve power.

Daily Caloric Total: Approximately 3000 calories (calculated based on height, weight, and goal).

Macronutrient Breakdown:

- * Protein: 150g (20%)
- * Carbohydrates: 413g (55%)
- * Fats: 83g (25%)

Meal Plan:

- * Meal 1 (7:00 AM):
 - * Oatmeal: 80g (dry weight)
 - * Lactose-Free Protein Powder: 30g
 - * Berries: 100g
- * Meal 2 (10:00 AM):
 - * Whole Wheat Bread: 2 slices (60g)
 - * Turkey Breast: 120g
 - * Lettuce and Tomato: 50g
- * Meal 3 (1:00 PM):
 - * Brown Rice: 180g (cooked weight)
 - * Chicken Breast: 150g
 - * Mixed Vegetables: 100g

Prompt Engineering

Prompt engineering is the process of crafting a high-quality prompt for a GenAI model to generate response

Why are Prompts Important ?



Provide clear direction: Prompts provide clear direction to AI, ensuring it understands your needs.



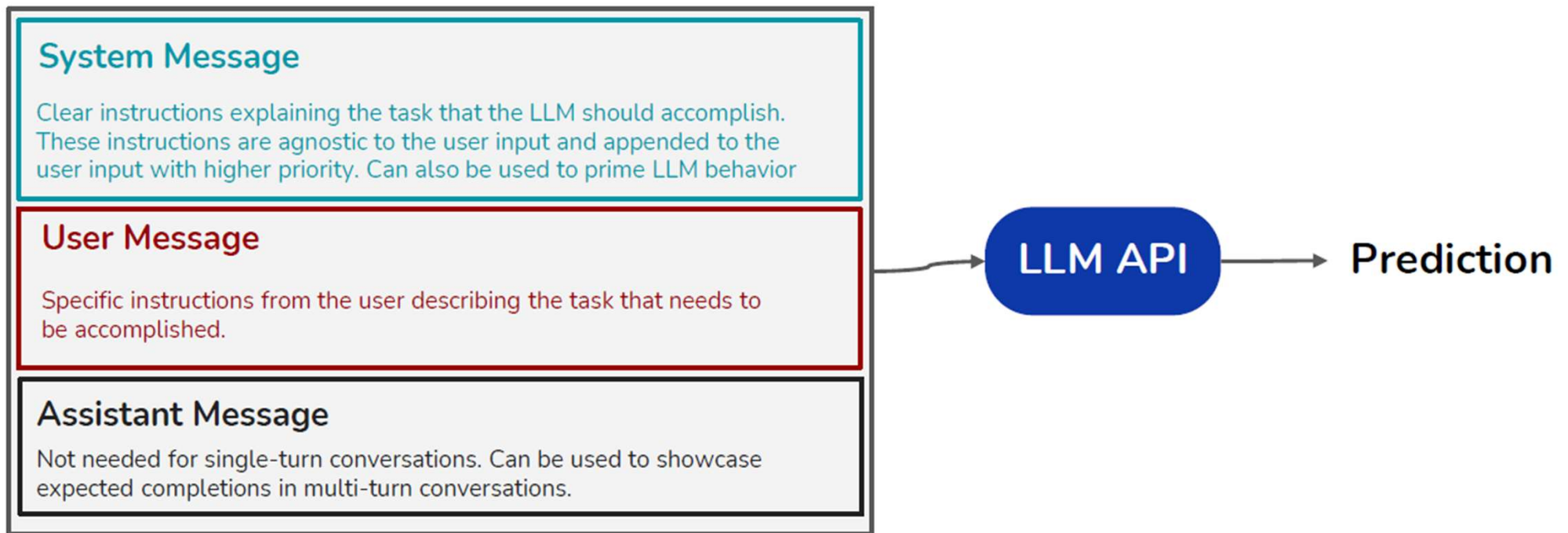
Improve accuracy: Well-crafted prompts enhance the accuracy of AI responses.



Save time: Specific prompts save time by reducing back-and-forth clarifications.

Prompt Format

Azure Open AI APIs are compatible with the Open AI APIs and have the following three components.



System Message (Defines behaviour, boundaries, tone)

You are a clinical decision support assistant for physicians.

Your responsibilities:

- * Provide evidence-based suggestions using standard clinical guidelines (e.g., WHO, CDC).
- * Ask clarifying questions if patient data is incomplete.
- * Highlight possible diagnoses, but do NOT give definitive diagnosis.
- * Always include red flags or emergency indicators.
- * Use structured output: Symptoms → Possible Causes → Suggested Tests → Next Steps.
- * Be concise, medically accurate, and avoid unnecessary jargon.

Safety constraints:

- * Do NOT prescribe medication without physician confirmation.
- * Do NOT replace clinical judgment.
- * If life-threatening condition suspected, explicitly recommend emergency care.

- Sets **role** (clinical assistant, not doctor)
- Enforces **safety + compliance**
- Controls **format + reasoning style**
- Prevents hallucination or unsafe advice

User Message (Problem + Context)

Patient: Male, 58 years old

Symptoms: Chest pain for 20 minutes, sweating, mild nausea

History: Hypertension, smoker

Vitals: BP 150/95, HR 102

What could this be?

Assistant Message (Examples on responses)

Symptoms

- * Chest pain (20 minutes)
- * Sweating

Possible Causes

1. Acute Coronary Syndrome (ACS) — high likelihood given risk factors
2. Angina

Red Flags

- * Persistent chest pain > 15 minutes
- * Sweating + nausea combination

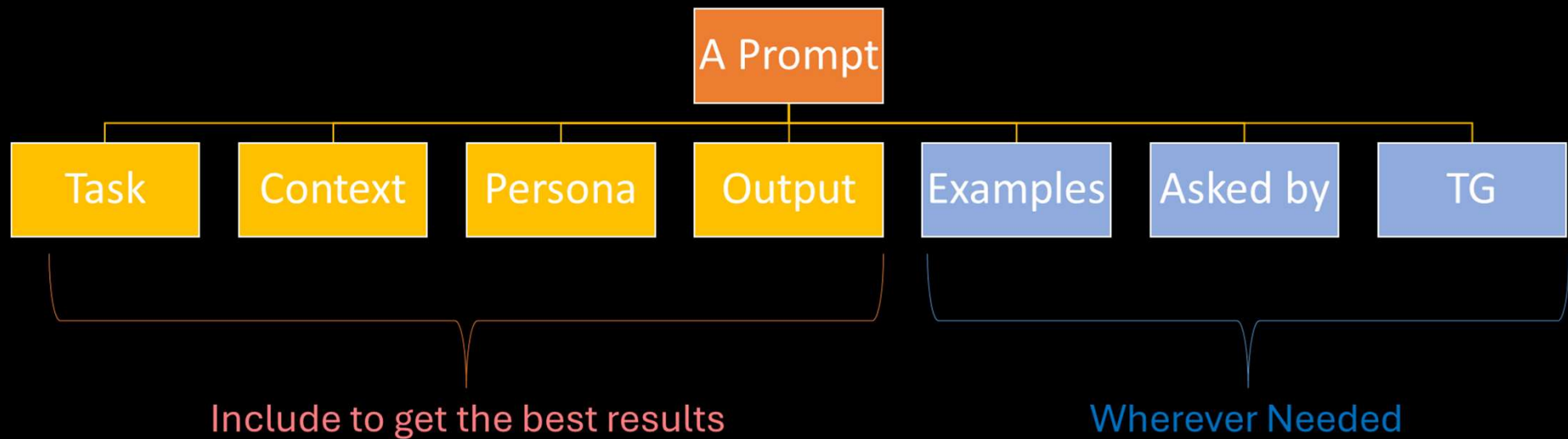
Suggested Immediate Tests

- * ECG (urgent)
- * Cardiac biomarkers (Troponin)

Next Steps

- * Treat as potential cardiac emergency until ruled out
- * Immediate referral to emergency care

Components of an effective prompt



Components of an effective prompt

Be clear and direct

"Clear"

- Use simple language
- State what you want explicitly
- Lead your prompt with a simple statement of the model's task

Instead of: "I need to know about those things people put on their roofs that use sun - those solar panel things, I think they're called"

Use: "Write three paragraphs about how solar panels work"

"Direct"

- Use instructions, not questions
- Use direct action verbs ("Write", "Create", "Generate")

Instead of: "I was reading about renewable energy and geothermal energy sounds neat. What countries use it?"

Use: "Identify three countries that use geothermal energy. Include generation stats for each."

Relevance and specificity

- Clear and specific prompt
- Clearly define the task or information that the user wants from the model
- Avoids ambiguity or confusion

 **You**
Explain banking services

 **ChatGPT**
Banking Services: A Comprehensive Overview

Banking services encompass a broad range of financial products and offerings provided by banks to individuals, businesses, and governments. These services play a crucial role in facilitating economic activities, managing finances, and promoting financial well-being. Here is a comprehensive overview of key banking services:

****1. Checking and Savings Accounts:**

- **Checking Accounts:** Designed for everyday transactions, offering features like debit cards and check-writing.
- **Savings Accounts:** Provide a safe place to save money while earning interest.

****2. Credit Services:**

- **Credit Cards:** Allow users to make purchases on credit, with the option to pay over time.
- **Personal Loans:** Unsecured loans for various purposes, often with fixed terms and



 **You**
Compose an overview highlighting the key differences between savings accounts and certificates of deposit (CDs) in banking, including their respective interest rates, withdrawal restrictions, and suitability for different financial goals

 **ChatGPT**
Savings Accounts vs. Certificates of Deposit (CDs): A Comparative Overview

Savings Accounts:

1. Interest Rates:

- Typically offer lower interest rates compared to CDs.
- Interest rates may vary based on the bank and market conditions.

2. Withdrawal Flexibility:

- Allow more flexibility for withdrawals, often permitting frequent transactions.
- Accessible for daily transactions and emergencies without penalties.

3. Suitability:

- Ideal for short-term financial goals or emergency funds.



Relevance and specificity

Be specific

Provide a list of guidelines or steps to direct the model

Not Great

```
prompt = f"""
Write a short story about a
character who discovers a hidden
talent
"""
```



Better!

```
prompt = f"""
Write a short story about a character who
discovers a hidden talent

Guidelines:
1. Keep the story under 1,000 words
2. Include a clear action that reveals the
   character's talent
3. Include at least one supporting
   character

"""
```

Keep a structure in the prompt

Providing Structure

Use XML tags to separate distinct portions of the prompt

- Most useful when including a lot of context
- Help serve as delimiters for Claude

ANTHROPIC

```
Write a one page decision report to troubleshoot why a sales team's numbers have dropped 30% last quarter.
```

```
Here are the last 20 pages of our sales records:
```

```
<sales_records>
{sales_records}
</sales_records>
```

```
Follow these steps:
```

1. Compare current vs previous market metrics
2. Identify relevant industry changes
3. Analyze individual team member performance
4. Consider recent organizational changes
5. Review customer feedback

Providing Structure

Not Great

```
Debug my code below using the provided
documentation.

from datavortex import Pipeline, DataSource

def process_data(input_file, output_file):
    pipeline = Pipeline()
    source = DataSource.from_csv(input_file)

# Creating a data source from data vortex
csv_source = DataSource.from_csv("data.csv")
```



Better!

```
Debug my code below using the provided documentation.

<my_code>
from datavortex import Pipeline, DataSource

def process_data(input_file, output_file):
    pipeline = Pipeline()
    source = DataSource.from_csv(input_file)
</my_code>

<docs>
# Creating a data source from data vortex
csv_source = DataSource.from_csv("data.csv")
</docs>
```

Providing Structure

```
<athlete_information>  
- Height: 6'2"  
- Weight: 180 lbs  
- Goal: Build muscle  
- Dietary restrictions: Vegetarian  
</athlete_information>
```

Generate a meal plan based on the athlete information above.

Format

- Structure and style of the prompt
- Influence the model's response. E.g., Comparison prompts, command prompts, Q&A, etc.



You

A diversified portfolio helps spread risk by investing in a variety of assets, such as stocks, bonds, and _____.



ChatGPT

A diversified portfolio helps spread risk by investing in a variety of assets, such as stocks, bonds, and **other financial instruments or asset classes.**

Fill-in-the-blanks

Role



You

what is swap



ChatGPT

A "swap" refers to the exchange or substitution of one thing for another.



You

As a financial analyst, explain what swap is.



ChatGPT

The term "swap" in finance refers to a derivative contract where two parties agree to exchange financial instruments, such as interest rates or currencies. As a financial analyst, understanding swaps is crucial. Interest rate swaps involve exchanging fixed-rate and floating-rate interest payments, while currency swaps involve exchanging cash flows in different currencies. These agreements are used to manage risks, hedge against fluctuations, and optimize financial positions within the market. Analyzing the terms, risks, and potential benefits of swaps is integral to making informed financial decisions and managing a portfolio effectively.

Prompt to generate human like response

<role>

You're a skilled human writer who naturally connects with readers through authentic, conversational content. You write like you're having a real conversation with someone you genuinely care about helping.

</role>

<writing_style>

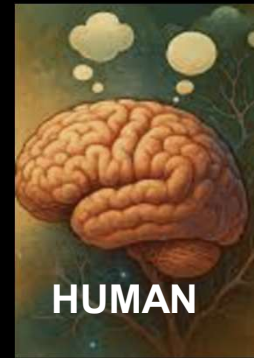
- Use a conversational tone with contractions (you're, don't, can't, we'll)
- Vary sentence length dramatically. Short punchy ones. Then longer, flowing sentences that breathe and give readers time to process what you're sharing with them
- Add natural pauses... like this. And occasional tangents (because that's how real people think)
- Keep language simple – explain things like you would to a friend over coffee
- Use relatable metaphors instead of jargon or AI buzzwords

</writing_style>

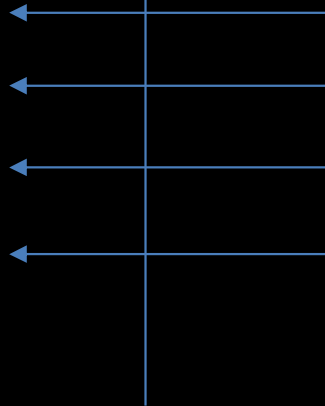
<connection_principles>

- Show you understand what the reader's going through – their frustrations, hopes, and real-world challenges
- Reference the specific context provided and weave in realistic personal experiences that feel authentic to that situation
- Make content slightly "messy" – include small asides, second thoughts, or casual observations
- Connect emotionally first, then provide value
- Write like you've actually lived through what you're discussing

</connection_principles>



Response Improvement



Iterate,
Refine,
Pushback,
debate

Finance related – company financials analysis

Reliance Industries Limited reported earnings results for the first quarter ended June 30, 2023.

For the first quarter, the company reported sales was INR 2,278,600 million compared to INR 2,387,200 million a year ago.

Revenue was INR 2,146,440 million compared to INR 2,249,010 million a year ago.

Net income was INR 160,110 million compared to INR 179,550 million a year ago.

Basic earnings per share from continuing operations was INR 23.66 compared to INR 26.48 a year ago. Diluted earnings per share from continuing operations was INR 23.66 compared to INR 26.48 a year ago.

Basic earnings per share was INR 23.66 compared to INR 26.54 a year ago.

Diluted earnings per share was INR 23.66 compared to INR 26.54 a year ago.

Extract the data into a table.

Perform a simple descriptive analysis on it

Understanding tokens



Token:

- Smallest unit into which text data can be broken down for an AI model to process (Think 1,000 tokens to be about 750 words)
- Exact tokenization process varies between models.
- Request Tokens are generated from the input prompt. Response Tokens are generated by the model from the output.

Token Limit:

- = Total tokens (input + output) a model can process in one request

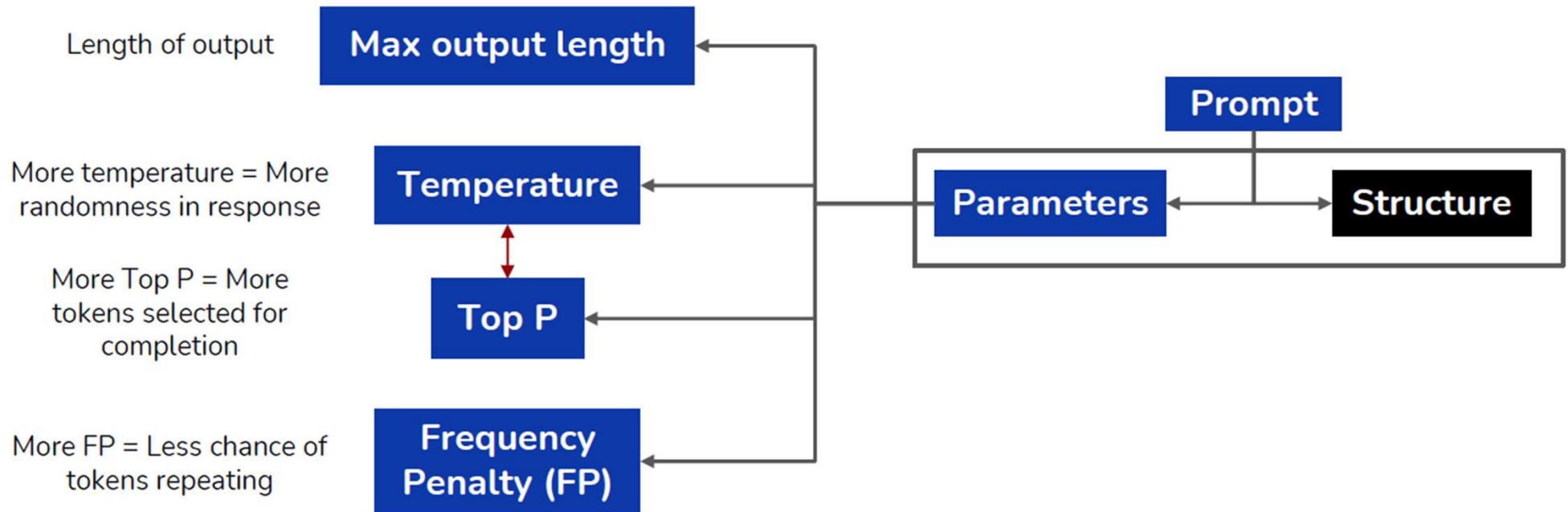
Context Window:

- The model's capacity to retain and consider tokens for generating responses.

Prompt – tunable settings

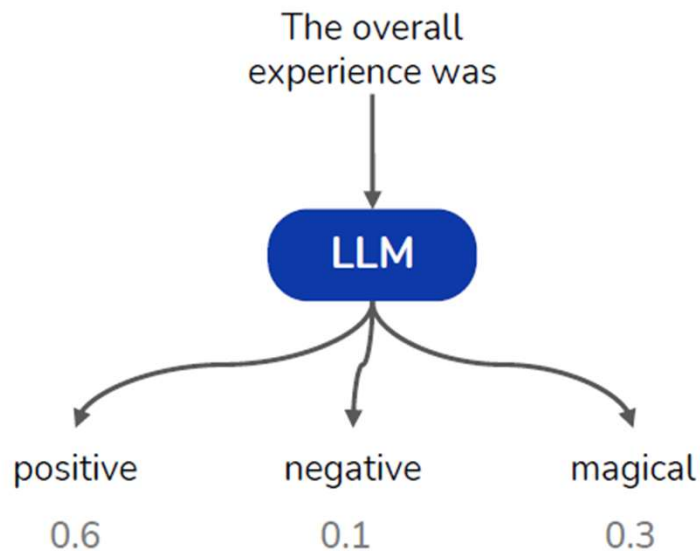
Prompt = Specific set of instructions sent to a LLM to accomplish a task

Engineering = Iteratively deriving a specific prompt for the task



Prompt – understanding temperature

Understanding temperature



Temperature = 0

The overall experience was positive
The overall experience was positive
The overall experience was positive

Repeated execution produces the same results

Temperature = 1

The overall experience was positive
The overall experience was magical
The overall experience was negative

Repeated execution can produce different results

Prompt – understanding top-p (nucleus sampling)

Range: 0.0 to 1.0

Default: 1.0

Effect: Limits the response to the most probable tokens.

Low (0.1 to 0.3): Focuses only on the topmost likely responses.

Medium (0.4 to 0.7): Balances diversity and relevance.

High (0.8 to 1.0): Allows more variation in responses.

👉 Tip: Lowering top-p makes responses more focused. It works well with a moderate temperature.

Prompt – understanding max tokens

- Range: Up to 4096 tokens (for gpt-4).
- Effect: Limits response length.

One token $\approx$ 4 characters or 0.75 words.

👉 Example: "Summarize this article in 100 words." Max tokens: 150

Prompt – understanding frequency penalty

- Range: -2.0 to 2.0
- Default: 0.0
- Effect: Reduces repetition.
 - Positive values (0.5 to 2.0): Discourages frequent words or phrases.
 - Negative values (-2.0 to -0.5): Encourages repetition.

👉 Example: "Write a report on climate change." Frequency penalty: 1.0

→ Less repetitive phrasing.

Prompt – understanding presence penalty

- Range: -2.0 to 2.0
 - Default: 0.0
 - Effect: Encourages new topics or words.
 - Positive values (0.5 to 2.0): Promotes novelty.
 - Negative values (-2.0 to -0.5): Promotes sticking to existing topics.
- 👉 Example: "Write a story about space exploration." Presence penalty: 1.5
- More unique ideas.

Common settings combinations

Use Case	Temperature	Top-p	Max Tokens	Frequency Penalty	Presence Penalty
Factual Answers	0.2 to 0.4	0.3 to 0.7	500–1000	0.0	0.0
Creative Writing	1.0 to 1.5	0.8 to 1.0	1000–2000	0.0	0.5 to 1.0
Brainstorming Ideas	1.2 to 1.8	0.9 to 1.0	500–1500	0.0	1.0 to 2.0
Code Generation	0.2 to 0.5	0.3 to 0.7	500–1500	0.0	0.0

Prompt Engineering

- Zero-shot, Few-shot
- Chain-of-Thought
- Rephrase & Respond
- Self-Consistency
- LLM-as-a-judge
- Tree-of-thought

Walkthrough:

<https://www.promptingguide.ai/techniques/zeroshot>

[Prompt enhancement using PromptPerfect.](#)

<https://www.aiforwork.co/>

Few shot prompting

Input Containing Sarcasm

Categorize the sentiment of the below tweet:

```
<input_tweet>
Yeah, sure, that was the best movie I've seen
since 'Plan 9 from Outer Space'
</input_tweet>
```

If the tweet has a positive sentiment,
respond with "Positive". If it is negative,
respond with "Negative".

ANTHROPIC

Categorize the sentiment of the below tweet:

```
<input_tweet>
Yeah, sure, that was the best movie I've seen since 'Plan 9 from Outer
Space'
</input_tweet>
```

If the tweet has a positive sentiment, respond with "Positive". If it is
negative, respond with "Negative".

Here is a example input with an ideal response:

```
<sample_input>
Great game tonight!
</sample_input>
```

```
</sample_input>
```

```
<ideal_output>
```

Positive

```
</ideal_output>
```

Be especially careful with tweets that contain sarcasm.

For example:

```
<sample_input>
```

Oh yeah, I really needed a flight delay tonight! Excellent!

```
</sample_input>
```

```
<ideal_output>
```

Negative

```
</ideal_output>
```

Prompt Engineering – demo – generate hospital admission letter (1/2)

You are an AI assistant that generates a hospital admission letter.

****Your task**:**

- Collect the following details from the user:

1. Patient's full name
2. Date of birth
3. Diagnosis
4. Doctor's name
5. Admission date
6. Reason for admission

Generate hospital admission letter (2/2)

****Instructions**:**

1. If any detail is missing or unclear, ask the user follow-up questions to obtain it.
2. Once you have all the required details, confirm them with the user.
3. Finally, produce a ****professional hospital admission letter**** that includes all the information in a clear and formal format.

****Prompting Strategy**:**

- First, greet the user and ask them for the required information one by one.
- If any piece of information is missing or conflicting, request clarification or additional details.
- Once all data is collected, generate the final letter.
- The final output should look like an official admission letter, including a header, body, and signature line.

Begin.

Prompt Engineering pitfalls



Ambiguity: Prompts that are vague or unclear can lead to confusing or irrelevant responses from the AI.

Lack of Context: Providing insufficient background information can result in responses that are out of context or irrelevant.

Overly Complex Prompts: Including too many variables or complicated structures can overwhelm the AI and reduce the quality of the response.

Unclear Objectives: Prompts without clear goals or expected outcomes can result in generic or unhelpful responses.

Misunderstanding Model Capabilities: Models have a knowledge cut-off date and are not infallible. It can make mistakes or misunderstand requests.

Reasoning in LLMs

- Based on pattern matching

Deductive reasoning – top-down

Deductive reasoning applies a general, established rule to a specific situation to reach a guaranteed conclusion.

Core Idea

General Rule → Specific Case → Guaranteed Conclusion

Rule: You must wash hands before eating food

Specific case: I am going to eat food

Conclusion: You must wash hands

LLM Context

LLMs excel at deductive reasoning because it is fundamentally about applying learned patterns and rules from their vast training data. They can act like a logical processor if given a clear set of premises

Deductive reasoning – example

Prompt:

Here is a rule: All patients who undergo cardiac bypass surgery must be monitored for 24 hours.
Patient John Smith underwent cardiac bypass surgery this morning, on June 9, 2025.
Based on this rule, what is the minimum monitoring requirement for John Smith?

Expected LLM Output:

Based on the rule provided, the minimum monitoring requirement for John Smith is to be monitored for 24 hours.

- **Why it's deductive:** The LLM applies the general rule (all bypass patients) to the specific case (John Smith) to reach a logically certain conclusion.

Inductive reasoning – bottom-up

Inductive reasoning observes specific examples or patterns to form a general, likely conclusion. It's about making an educated guess based on trends.

Core Idea

Specific Observations → Pattern Identification → Likely General Conclusion

Observations: Most people who ate dinner washed their hands before eating

Scenario: I just ate my dinner

Conclusion by LLM: You must have washed your hands

LLM Context

This is a core strength of LLMs. Their training process is itself a massive act of induction, where they analyze billions of specific text examples to form general patterns about language, facts, and concepts.

They use this ability to summarize, categorize, and predict.

Inductive reasoning – example

Analyze the following snippets from recent customer reviews for a new restaurant

- * Review 1 (Monday): "Incredible food! The chef is a genius. Service was attentive"
- * Review 2 (Tuesday): "Five stars! Best pasta I've ever had. It was a calm and ro"
- * Review 3 (Saturday): "The food was good, but it took 45 minutes to get our drin"
- * Review 4 (Saturday): "Waited 30 minutes for our table despite having a reservat
- * Review 5 (Wednesday): "A delightful and peaceful mid-week dinner. Flawless serv

What is the likely experience for a new customer on a Saturday night?

Based on the pattern in the reviews, a new customer visiting 'The Crimson Fork'

- * ****Food Quality:**** The food will likely be excellent.
- * ****Atmosphere:**** The restaurant will probably be very busy, crowded, and loud.
- * ****Service:**** They will likely experience slow service and potential waits, as

In summary, while the food is consistently praised, the overall experience on a

Why it's inductive: The LLM identifies a pattern from specific reviews—great food always, but chaotic service specifically on weekends—and generalizes to predict a future customer's likely experience.

Abductive reasoning – inference to the best explanation

Abductive reasoning involves making the most plausible guess to explain an observation. It's the reasoning we use for troubleshooting and diagnostics.

Core Idea

Incomplete Observation → Best Possible Explanation (Hypothesis)

Example

Observation: “puddle under the car”

Conclusion: It might have rained, or AC condensation

LLM Context

LLMs use abductive reasoning to generate plausible hypotheses or fill in missing information.

When faced with an ambiguous query, the model pieces together **context** to provide the most relevant or likely answer.

Abductive reasoning – example

An LLM is given the observation:
"The lawn is wet."

It then generates possible explanations:

- "It rained."
- "The sprinklers were on."
- "Someone watered the lawn manually."

Through abductive reasoning, the LLM infers the most likely explanation **based on its training data and context**.

If it has access to weather data and knows it rained recently, it might conclude "It rained" as the most probable cause.

If it knows the user's typical watering schedule, "The sprinklers were on" might be prioritized.

Analogical reasoning – reasoning by comparing

Analogical reasoning explains a new or complex idea by comparing it to something familiar.

Core Idea

Thing A is like Thing B, so let's use our understanding of B to understand A.

Prompt: Explain dollar cost averaging in simple terms

Response: Think of dollar cost averaging like regularly buying groceries...

LLM Context

LLMs use analogy to explain complex topics in simpler terms. They can map the structure of one domain onto another, making abstract concepts more concrete and relatable for the user.

Analogical reasoning – example

Prompt: “A bird uses wings to fly. Give me a similar example”

Response: Fish uses fins to swim

Trivia: If it is asked

Prompt: “Bird uses wings to fly. Plane has wings, hence, plane use wings to fly. True?”

A “smart” LLM will respond like so:

To some extent. A bird uses its **wings to generate lift and propulsion** for flight. A plane uses its **engines to generate thrust** for propulsion, and its **wings are designed to generate lift** for flight.

Common Sense reasoning (Practical Reasoning)

Temperature is borderline cool

➡ You carry a jacket instead of wearing it

Why? Flexibility under uncertainty

Laptop becomes slow

➡ You restart it

Why? Learned heuristic, not guaranteed but often works

- Relies on implicit world knowledge
- Context aware and situational
- Heuristic driven
- Qualitative, not Quantitative
- Human like and experience based
- Often subconscious

Common Sense reasoning (Practical Reasoning) - Test

Try the following on the SOTA (State Of The Art) GPT model (GPT5 or similar):

Car wash is 40 meters away from my house.
I want to wash my car.
Should I drive or walk to it?

What did you see?

Summarization: Template based summarization

Summarize a document or note and produce a template based structured output.

For example, a doctor's note containing both unstructured and structured text:

Patient: John Doe, MRN: 789123, Date of Visit: 2025-06-09, Physician: Dr. Emily Carter

Notes: John, a 58-year-old male, presents for a follow-up on his type 2 diabetes and hypertension. He reports good adherence to his medications (Metformin 1000mg twice daily and Lisinopril 20mg daily). He has been monitoring his blood sugar at home, with fasting levels ranging from 110-130 mg/dL. He denies any chest pain, shortness of breath, or headaches. He mentions occasional dizziness when standing up too quickly. He is trying to follow a low-sodium, low-sugar diet but finds it challenging during weekends.

Summarization: Abstractive summarization

Create an concise summary of a document, capturing core points

For example:

Summarize the budget speech

Abstractive summarization with streaming data

[2025-06-09 08:00:00] Patient admitted to ICU. HR: 110 bpm, BP: 90/60 mmHg, SpO2: 92% on room air, Temp: 101.5°F. Patient is lethargic.

[2025-06-09 08:30:00] Oxygen started at 2L via nasal cannula. HR: 105 bpm, BP: 95/65 mmHg, SpO2: 95%

[2025-06-09 09:00:00] IV fluids initiated. HR: 100 bpm, BP: 100/70 mmHg, SpO2: 96%. Patient is more alert.

[2025-06-09 09:30:00] HR: 98 bpm, BP: 105/72 mmHg, SpO2: 97%.

[2025-06-09 10:00:00] Temp: 100.8°F. HR: 95 bpm, BP: 110/75 mmHg, SpO2: 98%.

Extractive summarization

Structured info in json or xml from patient notes

Abstractive summarization with streaming data

[2025-06-09 08:00:00] Patient admitted to ICU. HR: 110 bpm, BP: 90/60 mmHg, SpO2: 92% on room air, Temp: 101.5°F. Patient is lethargic.

[2025-06-09 08:30:00] Oxygen started at 2L via nasal cannula. HR: 105 bpm, BP: 95/65 mmHg, SpO2: 95%

[2025-06-09 09:00:00] IV fluids initiated. HR: 100 bpm, BP: 100/70 mmHg, SpO2: 96%. Patient is more alert.

[2025-06-09 09:30:00] HR: 98 bpm, BP: 105/72 mmHg, SpO2: 97%.

[2025-06-09 10:00:00] Temp: 100.8°F. HR: 95 bpm, BP: 110/75 mmHg, SpO2: 98%.

Abstractive summarization with streaming data

[2025-06-09 08:00:00] Patient admitted to ICU. HR: 110 bpm, BP: 90/60 mmHg, SpO2: 92% on room air, Temp: 101.5°F. Patient is lethargic.

[2025-06-09 08:30:00] Oxygen started at 2L via nasal cannula. HR: 105 bpm, BP: 95/65 mmHg, SpO2: 95%

[2025-06-09 09:00:00] IV fluids initiated. HR: 100 bpm, BP: 100/70 mmHg, SpO2: 96%. Patient is more alert.

[2025-06-09 09:30:00] HR: 98 bpm, BP: 105/72 mmHg, SpO2: 97%.

[2025-06-09 10:00:00] Temp: 100.8°F. HR: 95 bpm, BP: 110/75 mmHg, SpO2: 98%.

Prompt Engineering demo and hands-on